INCH-POUND
MIL-C-17361/13A(SH)
13 March 1995
SUPERSEDING
MIL-C-17361/13(SH)
10 March 1989

MILITARY SPECIFICATION SHEET

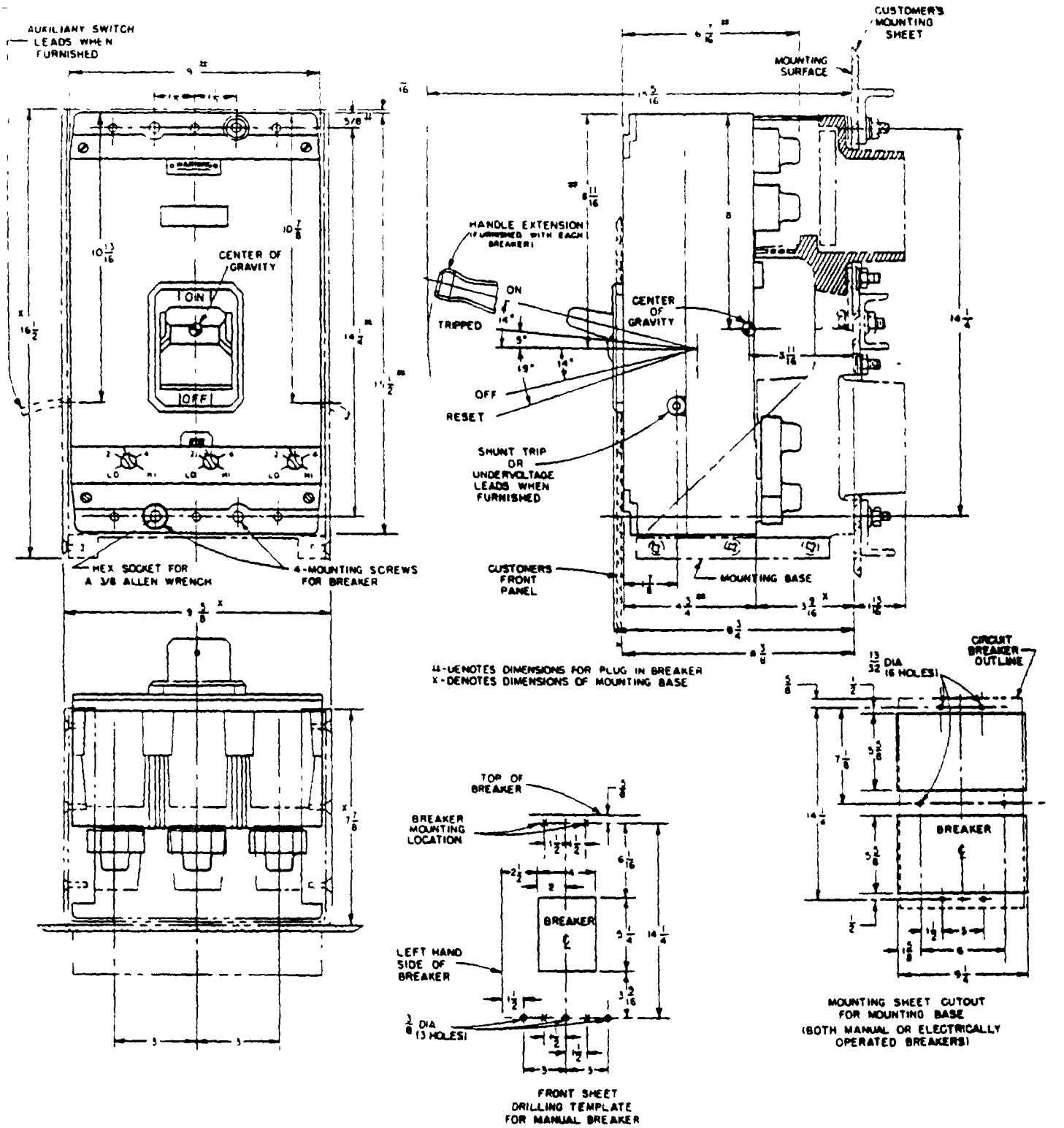
CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPE AQB - A1600, AQB - A1604, AND NQB - A1600

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation MIL-C-17361

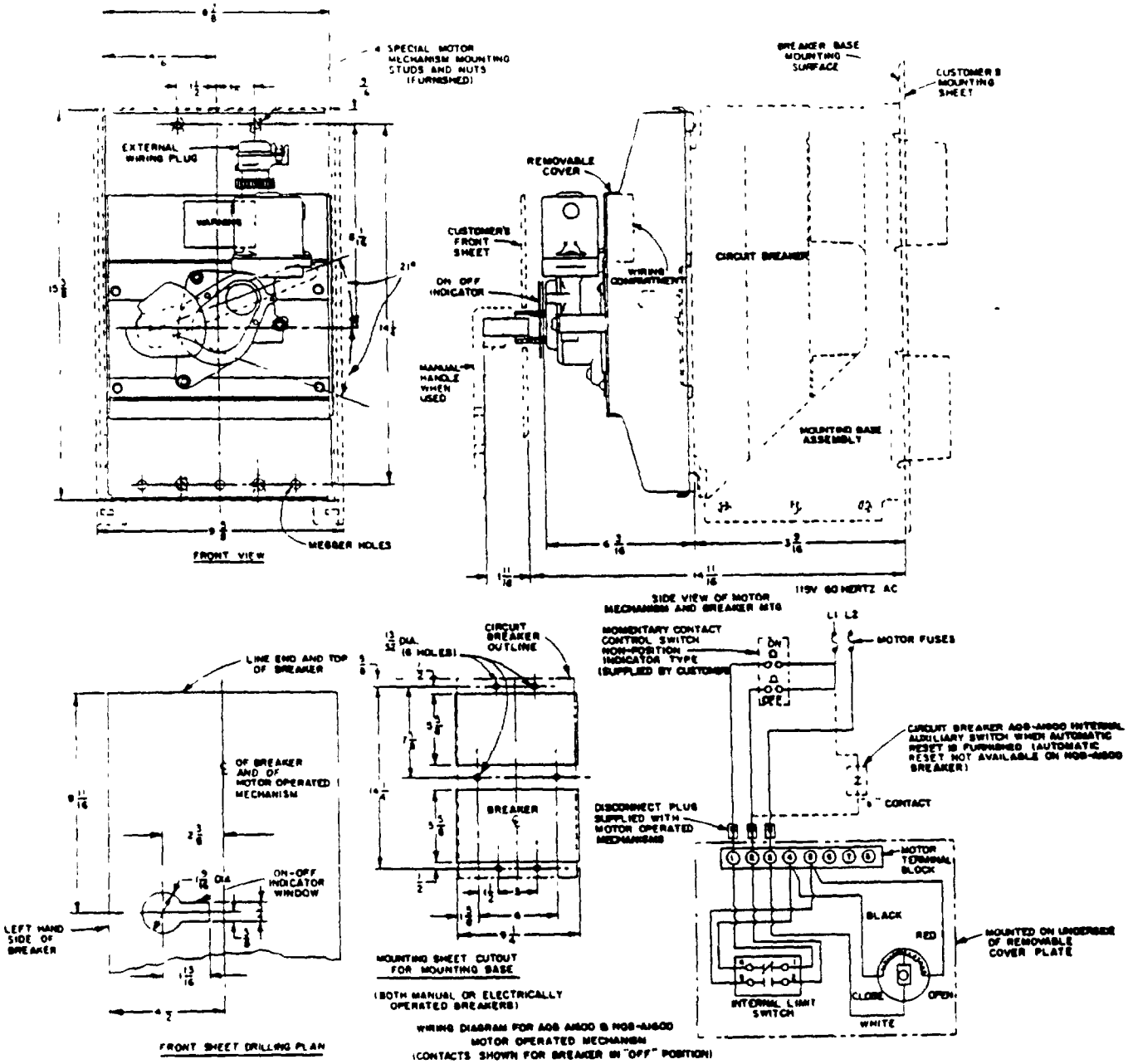
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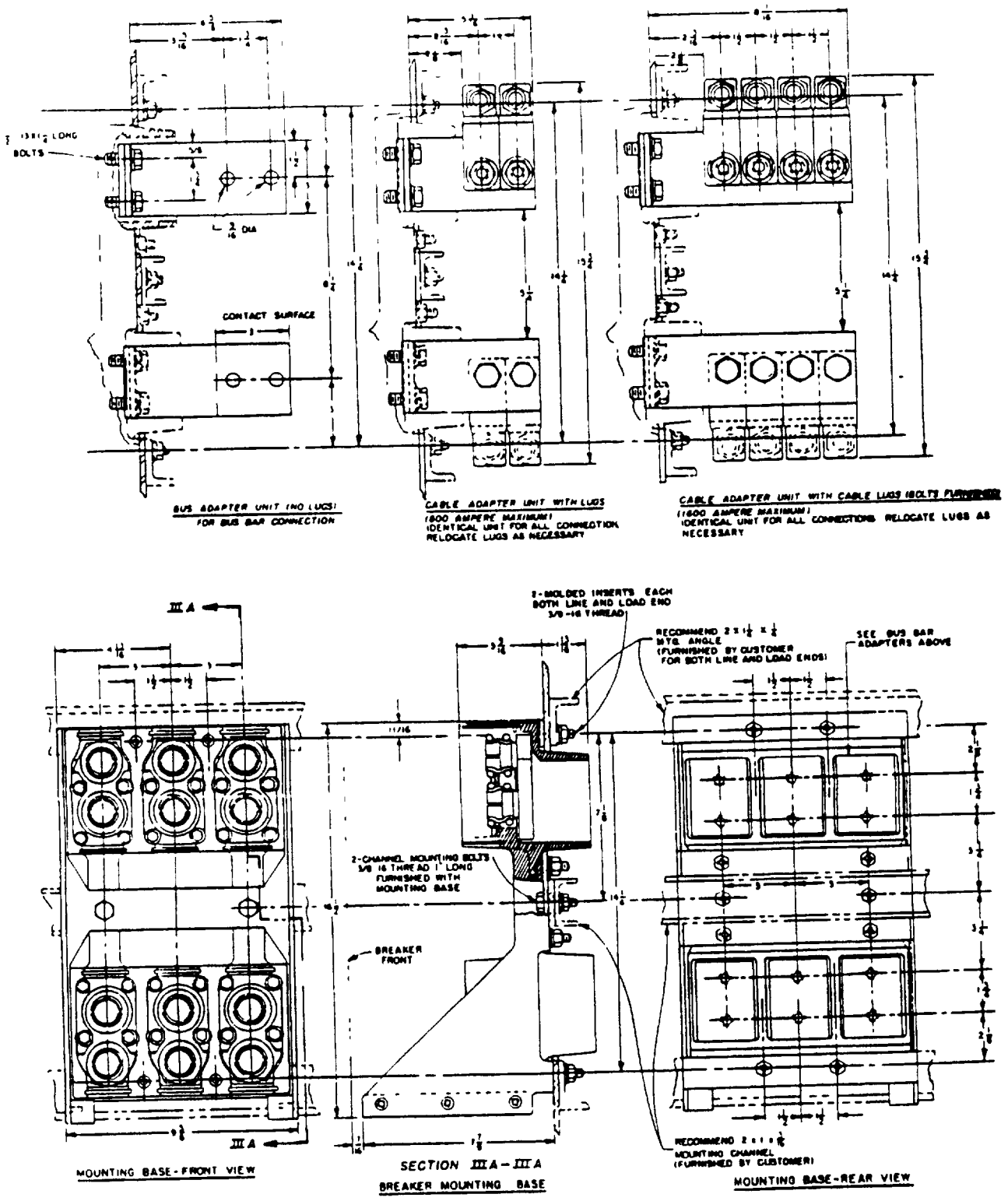
* FIGURE 1. Circuit breaker type AQB-A1600, AQB-1604, and NQB-A1600 outline and mounting dimensions.

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★ FIGURE 1. Circuit breaker type AQB-A1600, AQB-1604, and NQB-A1600 outline and mounting dimensions continued

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* FIGURE 1. Circuit breaker type AQB-A1600, AQB-1604, and NQB-A1600 outline and mounting dimensions continued.

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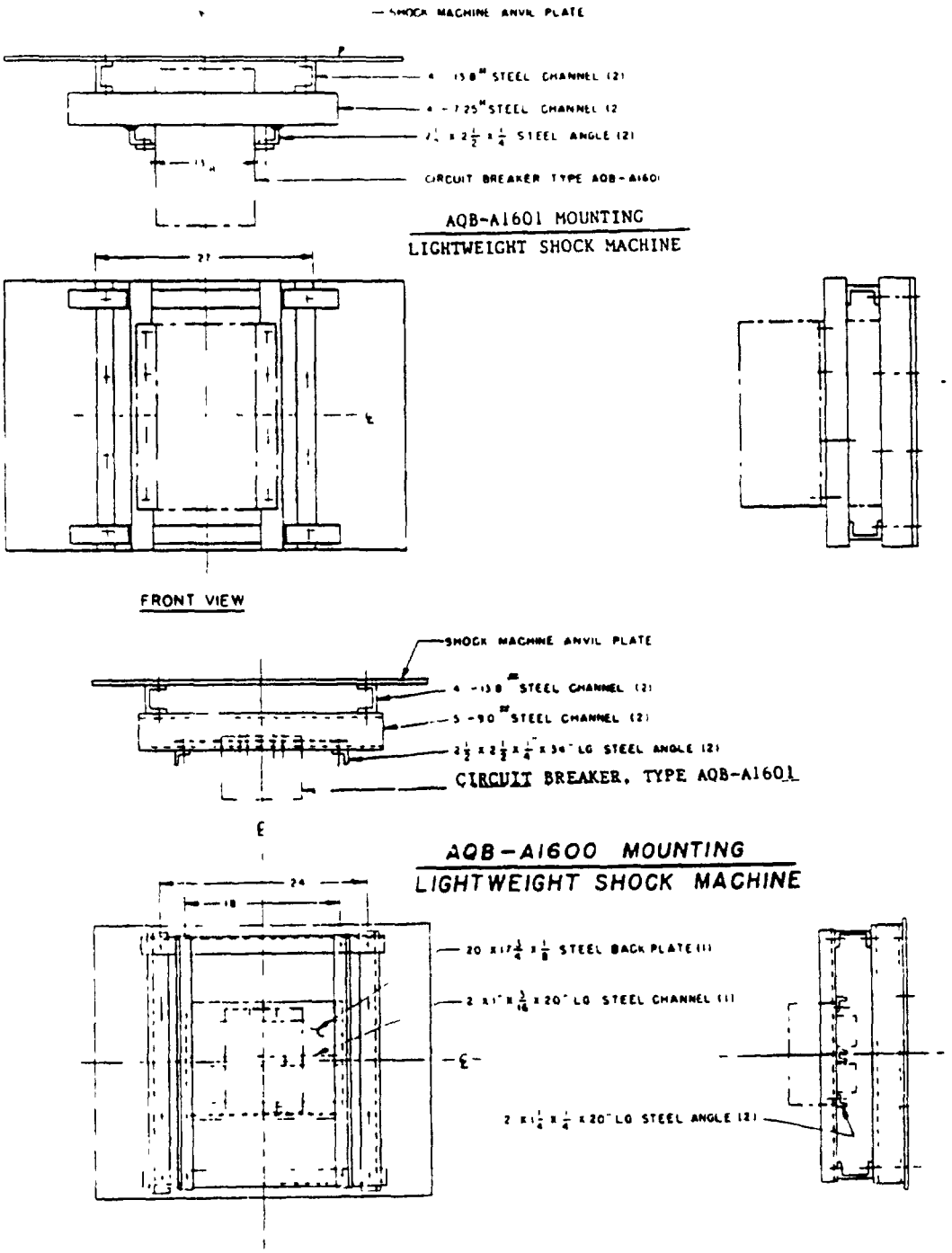


FIGURE 2. Standard mounting fixtures for shock testing of type AQB-A1600 and AQB-A1604 circuit breakers.

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REQUIREMENTS

Dimensions See figure 1
Maximum rated continuous current 1,600 amperes
Voltage and frequency 500 Vac, 60 Hz
Interrupting rating 75,000 amperes
Peak let-through current Not applicable
Trip unit Interchangeable
Protection (for trip units type AQB-A1600 only)
Trip unit rating (amperes) See table I
LTD and adjustable instantaneous, all 3 poles
LTD calibration See table II
Instantaneous calibration See table I
Protection (for trip units type AQB-A1604 only)
Frame rating 1600 amperes (ac only, no dc rating)
Continuous current settings (CCS) 600, 800, 1000, 1200, 1400, and 1600 amperes
LTD, STD, and Instantaneous, all 3 poles
LTD calibration See table IIa
STD calibration The short time delay pickup settings shall be 2X, 3X, 4X, 6X, 8X, and 10X the continuous current settings The short time delay "delay settings" shall be as specified in Table IIb
Instantaneous calibration The instantaneous pickup settings shall be 3X, 4X, 5X, 6X, 8X, 10X, 12X, and 15X the frame rating Production units shall be calibrated at the 8X setting. Opening time shall be 0 025 seconds or less when tested at 3X the instantaneous pickup (for qualification tests only, not for routine calibration)
Time characteristics curves (TCC) TCC shall be developed and provided Test points listed for LTD, STD, and Instantaneous shall comply with the TCC
Current limiting fuse Not applicable
Type of connections Back (removable mtg).
Electric operator Available
Undervoltage trip or shunt trip Either available
Auxiliary switch Four (a) contacts, four (b) contacts
Handle lock Not available
Handle boot Not available
Shock test See figure 2
Electrical test The test may be conducted in groups of 100 load operations
The rate of operation shall be 4 operations per minute for a total of 4000 close-open operations
Temperature rise tests See table III
Mechanical tests Mechanical (no-load) operations may be conducted between groups of load operations The circuit breaker shall be subjected to 1000 close-open operations

Note NQB circuit breakers shall consist of an AQB type without trip unit and a modification kit of metallic parts

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TABLE I Adjustable instantaneous trip settings amperes/maximum time 1/

Element rating 2/	LO 4/	2	3	4	HI	Seconds 3/
	Ac	Ac	Ac	Ac	Ac	60 Hz ac
600Y	2000	3000	4000	5000	6000	
600YM	4000	6000	8000	10000	12000	
800Y	2000	3000	4000	5000	6000	
1000Y	2000	3000	4000	5000	6000	0 032
1200Y	2000	3000	4000	5000	6000	
1400Y	4000	6000	8000	10000	12000	
1600Y	4000	6000	8000	10000	12000	

1/ The tolerance for each ampere setting shall not exceed plus or minus 10 percent

2/ Identification marking of trip units
Numerals indicate continuous current rating of the specific trip unit
Letter "Y" denotes that the trip units are applicable to AQB-A1600
Letter "M" denotes calibration for applications having a high inrush current

3/ Time shall be measured at three times the pick-up current of instantaneous setting For qualification tests only, not for routine calibration

TABLE II Tripping characteristics - long time delay (for trip units type AQB-A1600 only)

LTD trip element percent current	Tripping time 60 Hz ac trip element
150	Not less than 1 hour
225	Less than 1 hour
600	Between 20 and 180 seconds

TABLE IIa Tripping characteristics - long time delay (for trip units type AQB-A1604 only)

LTD in Multiples of CCS	Tripping Time 60 Hz ac
1 35	Not less than 1 hour
1 65	Less than 1 hour
4 0	Between 35 and 55 seconds
6 0	Between 20 and 32 seconds

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TABLE IIb Adjustable Short-Time Delay "Delay Settings" in Seconds (for trip units type AQB-A1604 only)

Short-Time Delay Pickup Setting	Short-Time Delay "Delay Setting"		
	1	2	3
All Settings	025 - 055 1/	065 - 095	105 - 135

1/ Time delay with 2X short-time delay pickup settings shall be 025 to 065 seconds

TABLE III Size of cable and bus bars for temperature rise tests.

Copper rating-amperes	Cable size	Copper bus size (inch)
1600	Four 400 MCM	3 x 1/4

Preparing activity
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